Filipino Text to Modern Baybayin Script in Image Form Using N-Gram and Modern Baybayin Script in Image Form to Filipino Text Using Convolutional Neural Network

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ABSTRACT
This study, Filipino Text to Modern Baybayin Script in Image Form Using N-Gram & Modern Baybayin Script in Image Form to Filipino Text Using Convolutional Neural Network aims to translate Filipino text to Baybayin Script Image and vice versa and help those people who are into this field. The study can translate Filipino Textby syllables, by word and by sentence. After conducting an experiment, the Filipino text to Modern Baybayin translation resulted to 95% accuracy, the Filipino text syllable to Modern Baybayin translation resulted to 98.6% while the Modern Baybayin Script to Filipino text reached 72% of accuracy and Modern Baybayin Script to Filipino text syllable reached 91% of accuracy.

General Terms
N-Gram, Convolutional Neural Network

Keywords
Bachelor of Science in Computer Science, Filipino Text, Kudlit, Modern Baybayin, N-Gram, Polytechnic University of the Philippines, Script.

1. INTRODUCTION
The word *baybayin*is a very old Filipino term that refers to the anci1ent script writing of the Filipinos prior to the Spanish colonization. Erroneously called *alibata,* Baybayin is a member of the Brahmic family used throughout South and Southeast Asia and was said to be used by ancient Filipinos as early as the 13th century long before the Spaniards came into the Philippines [1]. Originating from the word *baybáy,* Baybayin literally means to "spell". Incorrectly known as "Alibata", baybayin has been a core part of our culture and heritage for centuries. It has been the soul interconnectivity of our ancestors through writing. To spell; rooted from the word Baybay, our ancestors used our very own writing script to sign documents, write poetries, communicate, and write letters to one another. It was a form of communication that we once utilized in our everyday lives and was very well respected by the explorers who came to our islands." [2]. Recently, a House Committee Bill entitled Baybayin Act (House Bill 1022) or the National Writing System Act was passed which seeks to install Baybayin as the "National Writing System of the Philippines" – declaring it so and "providing for its promotion, protection, preservation and conservation". This declaration, consonant with the declared "policy of the state to inculcate, propagate and preserve the cultural heritage and treasures of the Philippines…" is necessary to make Baybayin "a tool for cultural and economic development to create consciousness, respect and pride for the legacies of Filipino cultural history, heritage and our authentic identity"[3].

Figure 1. Modern Baybayin Script

2. RELATED WORKS

N-GRAM

N-gram models are probabilistic models that assign probabilities on the "next" word in a sequence, given the n-1 previous words. The power of n-gram models was limited in natural language processing due to the fact that they cannot model deep dependencies, that is, the nth word only depends on the n-1 previous words; but they perform well on simple statistical demonstrations. Using a technique called N-gram analysis, it is possibly to very quickly extract sequences of characters from text and calculate the frequency of occurrences with the text [4].The application of n gram can be use in given the probabilities of a sentence it can determine the likelihood of an automated machine translation being correct, it could predict the next most likely word to occur in a sentence, it could automatically generate text from speech, automate spelling correction, or determine the relative sentiment of a piece of text [5].

CONVOLUTIONAL NEURAL NETWORK

In neural networks, Convolutional Neural Network (ConvNet or CNN) is one of the main categories for image recognitions and images classifications and areas such as objects detections and face recognition are some of the fields where CNNs are widely used. CNN image classifications works by taking an input image, processes it and classifies it under certain categories (Eg., Dog, Cat, Tiger, Lion). Computers sees an input image as array of pixels and it depends on the image resolution. Based on the image resolution, it will see h x w x d(h = Height, w = Width, d = Dimension). Eg., An image of 6 x 6 x 3 array of matrix of RGB (Red, Green, Blue) (3 refers to RGB values) and an image of 4 x 4 x 1 array of matrix of grayscale image. Technically, deep learning CNN models to train and test, each input image will pass it through a series of convolution layers with filters (Kernals), Pooling, fully connected layers (FC) and apply Softmax function to classify an object with probabilistic values between 0 and 1. The below figure is a complete flow of CNN to process an input image and classifies the objects based on values [6].

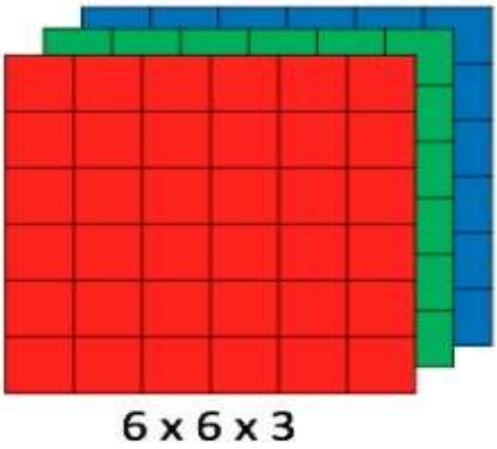

Figure 2. Array of RGB

SYLLABICATION

The division and partition of a word into syllables, in written or spoken form is called syllabication or syllabification. For most countries, the way that the syllables are uttered is also the basis of its syllabication once turned into its written form. But, potentially due to the weak connection between the letters in the spelling and the way that it is said, written syllabification is mostly based on morphological or etymological principles instead of phonetic concept [7]. Fern, a reading specialist in an elementary school often have students that struggle with decoding, or the area of reading involves sounding out words on a page. According to Fern there four important and most common rules to remember when decoding: (1) In a word, when a vowel is in between two consonants, the syllables should be segmented between the consonants. A good example of this rule include un/der, san/dy, or pig/let. (2) When two or more consonants is present in a word, the syllables should be divided so that the blends stay intact. Examples application of this second rule includes en/trance, mon/ster, or sub/tract. (3) In a word when two vowels have a consonant in between, the word should be divided after the first vowel, before the subsequent consonant. Some examples are mi/nus, a/ni/mal, and cu/rate. (4) If a word contains a double consonant, the syllables should be divided between the two consonants. Examples include hap/py, lit/tle, and diz/zy. [8]. In speaking the Filipino language, words are syllabicated or broken down by syllable in a slightly different way than in English. The process of how you break apart syllables when uttering the word/s has a significant impact on your accent and how clearly you speak and convey the language. The syllabication of Filipino words by syllables pertains to one unit of sound that forms the whole or part of a word. In Filipino, the word pantig is equivalent to the word syllable and pagpapantig pertains to the word syllabication. When syllabicating words, syllables are separated by a middle dot called the interpunct (·) which is also called the interpoint, mid dot, or centered dot. A syllable may be comprised of one vowel alone but a consonant needs a vowel to make a syllable. Syllables with two or more letters can have only one vowel. However, a syllable can have one or more consonants. The Filipino word for vowel is patinig (symbol: P) and the word for consonant is katinig (symbol: K) [9].

Table 1. Different forms of Filipino Syllable and Example

Syllable Form	Examples (in bold)
P	aka (a·ka), edad (e·dad), i·sa (i·sa), noo (no·o), oso (o·so), ubo (u·bo)
KP	ba·ka (ba·ka), dugo (du·go), relo (re·lo), si·la (si·la)
PK	akda (ak·da), isda (is·da), ugnay (ug·nay), uod (u·od)
KPK	aklat (aklat), bundok (bun·dok), dahon (da·hon), isip (i·sip)
KKP	bloke (blo·ke), braso (bra·so), droga (dro·ga), gripo (gri·po), plato (pla·to)
PKK	abstrak (abs·trak), ekstra (eks·tra)
KPKK	disk, fern, gibmefayb (gib·me·fayb), iskeyt (is·keyt), isport (is·port), keyk, kontest (kon·test), nars, test
KKPK	blangko (blang·ko), brilyo (bril·yo), dyip, dyakpot (dyak·pot), ekspres (eks·pres), ispred (is·pred), isprey (is·prey), kyut, plastik (plas·tik), playwud (play·wud), prinsesa (prin·se·sa), swak, trak, tren, tsek, tres, trangkaso (trang·ka·so), tsamba (tsam·ba), tsismis (tsis·mis)
KKPKK	bleyd, breyk, breyslet (breys·let), treynta (treyn·ta), tsart
KKPKKK	shorts

IMAGE TO TEXT TRANSLATION

There are some study that focuses for the translation of image of any writing system and to convert to text. In order to easily understand and teach other form of languages many researchers came up to an idea of translating words or writing systems. As stated in the thesis paper from the School of Engineering Technology West Bengal University of Technology, India, entitled, An Algorithm for Japanese Character Recognition, they proposed a geometry-topology based algorithm for Japanese Hiragana character recognition. The values of quantities both for the target and the template images are computed and math is made with the character having the minimum Euclidean distance with the average accuracy of 94.1% [10]. Another research focused on Thai Characters, according to the thesis paper from Stanford University, USA, entitled Recognition of Thai Characters and Text from Document Templates they created an image processing system that takes in an input image of a Thai text document, since it is an image processing kind of system it needs a phone camera. Taken from a phone camera and outputs a matching between each Thai character detected in the document to a particular Thai template character. The output could then be used to input into translation software to translate the Thai sentences out automatically without the need of a human translator [11]. Using XOR + Resize + WDS algorithm is the most favorable result with the 91% for clean (same font) 80% for clean (different font), 95% for test (same font), 83% for test (different font). On another research that focuses on hieroglyphics writing from the University of Amsterdam, Netherlands, entitled Automatic Egyptian Hieroglyph Recognition by Retrieving Images as Texts, they proposed an approach for automatically recognizing ancient Egyptian hieroglyph from photographs and segmented a large collection of nearly 4,000 hieroglyphs. They determine the reading order and subsequently evaluate 5 visual descriptors in 3 different matching schemes to evaluate visual hieroglyph recognition and with correctly annotated of hieroglyphics of 85.5 % automatic detection according to Pascal VOC overlap criteria [12]. On a similar research that proposed an image based document translation using an error correction model to correct those words who are recognized incorrectly from OCR output. The research is according to the writings from Computer Science Camegie Mellon University, USA, entitled Enhancing Image-based Arabic Document Translation Using a Noisy Correction Model. They train their correction model from synthetic data with different fonts and sizes to simulate real world situations and further enhance their recognition with bigrams to improve their word segmentation error correction. For instance, in an experiment using Arabic Transparent Font, the BLEU score increases from 18.70 to 33.47 with the use of their noisy channel model [13]. On a different approach, there are also researchers that studied the stenography writing translation into text, according to a thesis paper from the Polytechnic University of the Philippines, Philippines, entitled Gregg Stenography Image-To-Text Conversion and Vice Versa, they presented a tool to recognize written Gregg Stenography from an image to be translated into English text using Artificial Neural Networks (ANN) for pattern recognition and Natural Language Processing (NLP) algorithms for conversion of English text into its corresponding Gregg Shorthand. The research concerns in finding the completeness and correctness of the system's output, in text-to-image and vice versa using the specificity and sensitivity as the statistical treatment. With a degree of correctness of 10.72778 and 11.06389 for the degree of completeness [14]. In accordance to the paper by Obaidet. al. entitled Handwritten Text Recognition System Based on Neural Network, handwritten text recognition is still an open research issue in the domain of Optical Character Recognition (OCR). Their paper proposes an efficient approach towards the development of handwritten text recognition systems. 3-layer Artificial Neural Network (ANN) is utilized in their Paper using supervised learning approach. The choice of optimal feature vectors greatly the accuracy of any text recognition system therefore bit map representation of input samples are utilized as feature vector. The feature vectors are first pre-processed and then applied to the ANN along with the generated target vectors; that are generated on the basis on input samples. 55 samples of each English alphabet are used as a ANN training process in order to make sure the general applicability of system towards new inputs. Two different learning algorithms are utilized in this paper. Additive image processing algorithms are also developed in order to deal with the multiple characters input in a single image, tilt image and rotated image. The trained system provides an average accuracy of more than 95 % with the unseen test image [15]. According to the paper by Prof. Dr. Klaus Schneider entitled Optical Character Recognition - A Combined ANN/HMM Approach. They presented a new OCR method for degraded text recognition by introducing a combined ANN/HMM OCR approach. The approach provides significantly better performance in comparison with state-of-the-art HMM based OCR methods and existing open source OCR systems. In addition, the thesis introduces novel applications of ANNs and HMMs for document image preprocessing and recognition of low resolution text. Furthermore, the thesis provides psychophysical experiments to determine the effect of letter permutation in visual word recognition of Latin and Cursive script languages. Mostly, previous attempts in combining the HMMs and ANNs were focused on applying ANNs as approximation of the probability density function or as a neural vector quantizer for HMMs. These methods either require combined NN/HMM training criteria [ECBGMZM11] or they use complex neural network architecture like time delay or space displacement neural networks [BLNB95]. The presented system achieves a 30% reduction in error rate as compared to Google's Tesseract OCR system [Smi13] and 43% reduction in error as compared to OCRopus OCR system [Bre08], which are the best open source OCR systems available today. The proposed method resulted in more than 95% script recognition accuracy on various multi-script documents at connected component level and 100%-page orientation detection accuracy for Urdu documents [16]. According to the paper from ICCC International Digital Design Invitation Exhibition presented by Recario, Reginald Neil C. et al. entitled, An Automated Reader Philippine Baybayin Scripting Image Processing Method. The description of their work is that there are more recent discoveries in the Philippine artifacts bearing the Baybayin scripture and with the institution of the National Script Act of 2011, they presented an automated reader for the Baybayin scripts. The system is meant to give significant impact in promoting and reviving the use and relevance of the Baybayin script. The system generally analyses an input image of a Baybayin script and outputs the equivalent syllables of the script. The system is independently tested using the Berthod and Maroy's Primitives and Freeman Chain Coding. They used independently implemented by means of OpenCv[17].

TEXT TO IMAGE TRANSLATION

There are some study that focuses for the translation of text to an image of writing system. From the thesis paper from DAV College, India, entitled N-Gram Machine Translation System for English Punjabi Transliteration, He make a system that is made in such a way that it first learns from existing examples stored in the database and then with the help of N-Gram approach he translate the proper nouns of English language into its equivalent Gurumukhi Script. With a accuracy of 97% for the training of 15,000 and test of 2000. The system is tested on the various names belongs to various regions and overall accuracy of the system is very good [18]. From another research paper from Polytechnic University of the Philippines, Philippines, entitled Filipino Text Baybayin Alphasyllabary Translator, the researcher developed a mobile-based translator of Filipino Text to Baybayin Character for people to easily understand the alphasyllabary system. The study is for people who are interested in studying and teaching baybayin alphasyllabary system. The researchers used n-gram as the algorithm in obtaining the corresponding syllabication rule of a word. The proponents tested the accuracy of syllabication in translating words and sentences in terms of affixation, cluster and diphthongs with an accuracy of 99.2% [19].

3. METHODOLOGY PROPOSED SYSTEM

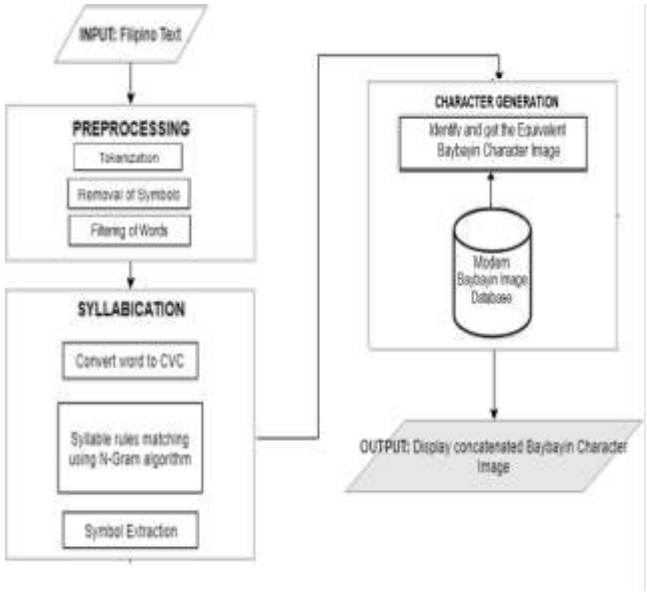

Figure 3. Filipino Text to Modern Baybayin System Architecture

A. Input Filipino Text – the phase where the proponents input Filipino text. .

B. Preprocessing – this process consists of tokenization which will segment words, sentences or phrases into set of one word and removal of other invalid symbol (‘@’, ‘_’, ‘|’, ‘#’, ‘$’, ‘%’, ‘^’, ‘&’, ‘=’, ‘+’, ‘*’, ‘<’, ‘>’, ‘`’, ‘~’, ‘[’, ‘]’, ‘(’, ‘)’, ‘{’, ‘}’, ‘:’, ‘;’, ‘“’, ‘‘’, ‘\’ ,’/’).

C. Syllabication – this process consists of converting the set of one word into CVC, wherein each letter will be converted into C if it’s Consonant or V if it’s Vowel. After converting, the N-gram algorithm will be used for syllable rules matching. The output of this process is the syllabicated word/s.

D. Character Generation – To select and identify the equivalent Modern Baybayin Character of each syllables and get the Modern Baybayin Character image from the database.

E. Output – this will show the concatenated equivalent Modern Baybayin Character Image of the syllables that will form the word/s.

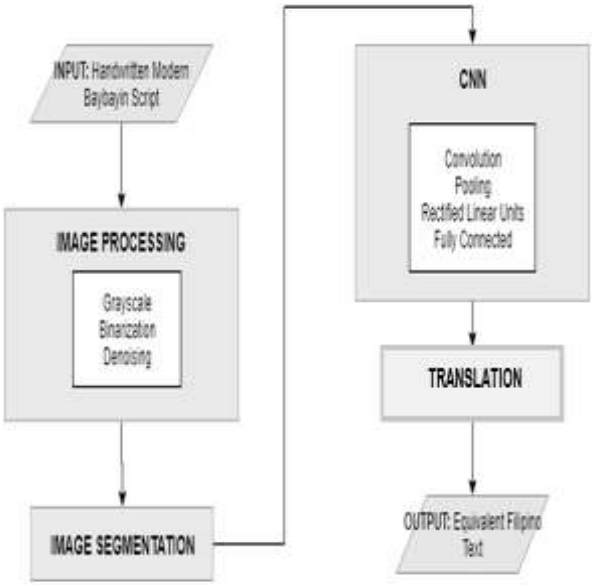

Figure 4. Modern Baybayin Script to Filipino Text System Architecture

A. Handwritten Modern BaybayinScript Image – This is the first phase of Modern Baybayin to text translation. The user or proponents will prepare a hand written Modern Baybayin Character/s written in a white clean bond paper using black pen.

B. Image Preprocessing – the scanned handwritten characters will now undergo preprocessing to improve the image data. The researchers used four different techniques for the preprocessing of the images. These four preprocessing techniques are the following:

- Grayscale - The process of converting the image into black-and-white in order to detect edges more accurately. For this are the proponents will be using Luminosity Method.
- Binarization – the process of converting the grayscale image into binary image. To address this process the proponents will be using Gatos Method.
- Denoising – the process of removing unnecessary noises from the image. For this part of the pre processing the proponents will be using Median Filter.

C. Image Segmentation – process of partitioning or dividing the image into multiple segments. For this process the proponents will be utilizing Horizontal Projection method.

D. CNN – after the features of the image have been extracted it will be fed to the

Convolutional Neural Network for classification before outputting the final result. Generally there are 4 layers inside the CNN (Convolutional Neural Network), and these are the following:

- Convolution – extract features from the input image.
- Pooling – reduces the dimensionality of each feature map.
- Rectified Linear Units (ReLU) - replaces all negative pixel values in the feature map by zero.
- Fully Connected Layers.-a traditional Multi Layer Perceptron that uses the features gathered for classifying the input image into various classes based on the training dataset.

E. Translation – after the image has been classified, it will now be translated according to its corresponding text.
F. Equivalent Filipino Text – the final output of the system from the handwritten modern baybayin script.

DATA GENERATION

- The researchers conducted an interview to an expert named, Mr. Nilo T. Boquirin a Teacher I from Panghulo National High School for them to have an insight about Modern Baybayin Characters/Script and also to gain knowledge that the researches will use to build this study.
- Researcher prepared 100 list of Filipino Text to be inputted in the translation of Filipino Text to Modern Baybayin in Image Form.
- Researcher prepared 100 list of Modern Baybayin Script to be scanned and inputted in the translation of Modern Baybayin in Image Form to Filipino Text.
- The researchers determined the methods, procedures, and techniques that will produce the expected output of the study.

4. RESULTS AND ANALYSIS

DATA ANALYSIS

To obtain the right accuracy and reliability in translating Filipino text to Modern Baybayin Characters/Script and vice versa.

Accuracy:

To calculate the system's accuracy, the data that gathered was examine using the system used.

ER = ((total - RC) / total)
RAP = (1 - ER) * 100%
Where:
ER = Error Rate
RAP = Recognition Accuracy Percentage
RC = Recognized Correctly
Total = Total number of Input

Reliability:

The formula to get the Reliability of this Research, the researchers use the Pearson R test.

$$r = \frac{N\Sigma xy - (\sum x)(\Sigma y)}{\sqrt{[N\Sigma x^2 - (\Sigma x)^2][N\Sigma y^2 - (\Sigma y)^2]}}$$

N = number of paired scores
Σxy = sum of the product of paired scores
Σx = sum of the product of x scores
Σy = sum of the product of y scores
Σx^2 = sum of squared x scores
y^2 =sum of product y scores

DISCUSSION

Table 2.Summary of Results of the System in terms of Filipino Text to Modern Baybayin

Category	Number of input	Accuracy Rate	Reliability Rate
Per Text	100	95%	100%
Per Syllable	4,250	98.6%	100%

Shows the summary of results of the system in terms of Filipino Text to Modern Baybayin Script translation with the input of 100 set of Filipino Text.The translation per text resulted to 95% Accuracy rate since there are some text that can't be translated by the developed system. And with the reliability of 100%. As per syllable, there were 4,250 syllable in the 100 Filipino Text and resulted to 98.6% accuracy and 100% reliability.

Table 3. Summary of the Results of the System in terms of Modern Baybayin Script into Filipino Text

Category	Number of input	Accuracy Rate	Reliability Rate
Per Script	100	72%	100%
Per Syllable	2,244	91%	100%

Shows the Summary of the Results of the System in terms of Modern Baybayin Script into Filipino Text. Overall, the translation per text resulted to 72% Accuracy rate and with the reliability of 100%. As per syllable, there were 2,244 syllable in the 100 Filipino Text and resulted to 91% accuracy rate and 100% reliability rate.

5. SUMMARY OF FINDINGS

Based from the summary of the findings for the translation of Filipino text to Modern Baybayin it is resulted to 95% Accuracy rate and 100% Reliability rate. As per syllable, 98.6% for the accuracy and 100% as well for the reliability because it resulted a consistent output just like the translation of Filipino Text.

In the translation of Modern Baybayin Script to Filipino text it is resulted to 72% accuracy rate and 100% reliability rate since it has a consistent result in the three trials conducted by the researchers. As per syllable, the accuracy rate reached 91% and 100% to reliability. The system shown consistency and stability in those testing being conducted, and so the developed system is proved to be accurate and reliable.

6. CONCLUSION

In this study, N-Gram and Convolutional Neural Network algorithm were being utilized to translate Filipino Text to Modern Baybayin Script and vice versa. Thorough analysis and simulation were done by the researchers to measure the accuracy and reliability of the developed system. The researchers conclude the following:

1. The researchers were able to prove that the developed system can translate Filipino Text to Modern Baybayin Script with the accuracy rate of 95%. The translation of Filipino Text Syllable to Modern Baybayin Script with the accuracy of 98.6% because the other "balbal words" cannot translated like charing, charot and cannot translated other foreign language. The researchers conclude that the developed system needs more improvement in terms of the translation of Modern Baybayin Script to Filipino Text Translation in terms of accuracy per text since the accuracy of Modern Baybayin Script to Filipino Text translation reacher 72% and 91% per syllable because there are times the system cannot validated all characters in the modern Baybayin image. The future researchers should feed more data accurately and maybe used different algorithm for the system to have a high accuracy.
2. The researchers also conclude that it already answered the statement of the problem to improve the accuracy of past research made by Pedroza et. Al with the accuracy of 99.61% and expand the scope by means of having a validation of Filipino Text, can translate Proper nouns and all kinds of letters since Modern Baybayin has 28 Script unlike the old one which is compose of 17 scripts only. It is resulted to 95% and 98.6% accuracy.

7. RECOMMENDATION

Based on the findings of the study, the researchers present the following recommendations for future work

1. Future researchers may develop a mobile version of this system. Because in acquired an image this prototype needed a third-party scanner to scan the image and input in the system for the purpose of time can lessen for acquiring an image, mobile phone have captured and automatically translated the image of Modern Baybayin into text.
2. Future researchers may also improve the Modern Baybayin image to text translation in terms of having an input. They may improve in terms of materials used when writing Modern Baybayin Script.
3. Future researchers may improve the Filipino text translation in terms of automatically check the grammar of the sentence or the text being input and automatically correct it, also the spelling of the wrong Filipino words.

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ABOUT THE AUTHORS

Andrea B. Alovera. She developed an early interest in computer when back when she was in high school. She spent his High School years at Jeremiah Christian Academy for four (4) years. She's a graduating student of Polytechnic University of the Philippines Sta., Mesa Manila with the degree of Bachelor of Science in Computer Science with specialization to Information System Development. During her college years, She is involved in 2 organizations: Association of Student for Computer Intelligence and Integrationand PUP College Y-Club affiliate by YMCA Manila. She is also knowledgeable in different programming language such as Java, COBOL, C and etc but her most favorite is Web programming language.

Greg Daniel N.Daisog. A graduate of Dr. Manuel S. Tansinsin Memorial School in in his elementary education and of Assumpta Academy in high school years, Greg Daniel Nierves Daisog is currently a graduating student of the Polytechnic University of the Philippines batch 2018-2019 with the course of Bachelor of Science in Computer Science with the specialization in Information Systems Development. A middle child among three siblings and has a knack for universal history and archaeology. An aspiring back-end web developer and COBOL programmer, he is pretty much excited to learn more about the IT industry and expand his technical knowledge in the world of programs.

Jericho G. Escarda. He studied primary education in Talolong Elementary School till his 4th Grade(2004-2009) and transferred in Manila in his 5th Grade. He continued his primary education in Santolan Elementary School for 2 years(2009-20011) and studied secondary education in Santolan High School for 4 years(2011-2015). He is a 4th year student and currently studying in Polytechnic University of the Philippines Academic Year 2018-2019 with a Bachelor Degree in Computer Science. He is currently living with his parents and 3 brothers in Santolan, Pasig City. He has also scholarship which aid his other financial problem. He is a Baptist and the church he is attending is the Sumulong Baptist Church. He is an introvert person because he doesn't want to talk as much as possible, the only time he wants to talk is when there is something important to talk about.

Benedick G. Gonzales. He the son of Helen G. Gonzales and Johnson R. Gonzales.His two brothers were John Patrick G. Gonzales and Jonel G. Gonzales and two sister named Jonnalyn G. Gonzales and Michelle G. Gonzales. He's a good child of his parent and he is intelligent and industrious. He studied in Polytechnic University of the Philippines and he takes a course Computer Science. In elementary he has a good honor on his study in Burgos Elementary school and joining in any drawing contest and in secondary he has been a SSG officer in Burgos National High School and a journalist and a cartoonist in The Greenstalk and have been joining in competition in AralingPanlipunan.

Aleta C. Fabregas. She is presently Assistant Professor IV in Polytechnic University of the Philippines. She obtained a bachelor's degree in Computer Data Processing Management and Master in Information Technology degree from Polytechnic University of the Philippines – Graduate School, Manila. Her field of specialization are Programming languages, System Analysis and Design, and Capstone. Has skills in data warehousing algorithms, database design, programming and information system security algorithms.